MIL-S-13352A(Ord)
2 February 1960
SUPERSEDING
MIL-S-13352(Ord)
29 March 1954

MILITARY SPECIFICATION

SIGHTUNITS: M34 SERIES

1. SCOPE

1.1 This specification covers a basic type of instrument used for direct or indirect fire for both elevation and azimuth of Rocket Launchers and Recoilless Rifles. Each sightunit is a composite instrument comprising the following component parts:

Elbow, Telescope: M62 Series
Adapter, Telescope: M9A1
Mount, Telescope: M79

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-S-12214 | Sightunits: M34, M34A1 and M34A2 Packaging of |
| MIL-M-13318 | Mount, Telescope: M79 |
| MIL-T-13321 | Telescope, Elbow: M62 (Series) |
| MIL-A-13325 | Adapter, Telescope: M9A1 |
| MIL-O-13830 | Optical Components for Fire Control Instruments: General Specification Governing the Manufacture, Assembly and Inspection of |
| MIL-F-13926 | Fire Control Materiel: General Specification Governing the Manufacture and Inspection of |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-130 | Identification Marking of U.S. Military Property |

FSC 1240

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DRAWINGS

Ordnance Corps

| | |
|---|---|
| F8244840 | Sightunit: M34A2, M34A2C |
| IEL-8244840 | Inspection Equipment List |
| LSQAP-8244840 | Supplemental Quality Assurance List |

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model.- At the beginning of production one sightunit, representative of production shall be made available to the Government to determine if the sightunit meets all requirements of this specification.

3.2 Materials.- Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.3 Design and construction.- The sightunit shall be manufactured in accordance with the drawings, applicable to the model sightunit specified in the contract, and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired optical and mechanical characteristics, performance and permanence to insure inherent quality and performance capabilities prior to final acceptance by the Government.

3.4 General specifications.- The contractor shall be responsible for adherence to, and compliance with the following requirements of Specifications MIL-O-13830 and MIL-F-13926.

3.4.1 MIL-O-13830

(a) Cleanliness
(b) Illuminated reticle

3.4.2 MIL-F-13926

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish painting and protective
(e) Workmanship



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3.5 Component items.- Prior to assembly as a sightunit, the following components shall have met the requirements of their respective specifications:

Mount Telescope: M79 - Specification MIL-M-13318
Telescope, Elbow: M62 (Series) - Specification MIL-T-13321
Adapter Telescope: M9A1 - Specification MIL-A-13325

3.6 Performance.

3.6.1 Orientation.- The sightunit, at standard ambient temperature (plus 60 degrees to 90 degrees Fahrenheit), shall be positioned with the dovetail axis of the mount vertical and the flat of the dovetail axis in a vertical plane.

3.6.2 Level vials.- The elevation level vial shall be centered. The cross level vial shall be centered within the width of a vial graduation line.

3.6.3 Horizontal alinement.- With the sightunit positioned in accordance with 3.6.1 and all graduations indexed at zero, and the line of sight of the telescope horizontal and coincident with a horizontal target line, the line of sight shall not deviate from the larget line by more than plus or minus 1.7 mils (total spread of 3.4 mils) when the adapter is rotated through 6400 mils by means of the azimuth micrometer knob.

3.6.4 Elevation jump.- Displacement between a line of sight through the telescope coincident with a horizontal target line, shall not exceed 0.5 mil when reversing the azimuth motion of the telescope by rotation of the azimuth micrometer knob.

3.6.5 Telescope position.- With the telescope vertical, the horizontal reticle line shall be in coincidence with a horizontal target line within one degree when the index of the telescope is in coincidence with the index on the holder assembly of the adapter.

3.7 Identification marking.- Identification marking unless otherwise specified shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor inspection.- Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.



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4.1.1 Contractor's quality assurance system.- The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies under this contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when it provides:

(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.
(b) Contractor's quality standards of acceptability.
(c) Procedure for continually evaluating product quality.
(d) Procedure for continually reporting evidence of actual product quality.
(e) Measures for preventing any significant departure of product quality from standards.

4.1.1.1 The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether or not certain procedures of the system provide equal assurance, the comparable procedure of this specification shall be used. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that might affect the degree of assurance required by this specification or other documents referenced herein.

4.1.2 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine whether the practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an affect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor will designate a responsible official or officials to whom the Government inspector will report such instances.

4.2 Production model.



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4.2.1 Unless otherwise specified in the contract, one unit taken from the production run shall be tested by the contractor under the surveillance of the Government.

4.2.1.1 These tests shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment designed or procured for use on the remaining items.

4.2.1.2 The model shall meet all the requirements of this specification using the inspection methods prescribed herein.

4.2.1.3 Disposition of rejected production model.- Should the production model fail to pass the test for Government acceptance, production of those components causing rejection will cease until the defective characteristics have been corrected in the production process. When corrections in the process have been made, a new production model shall be completely inspected by the contractor, and witnessed by the Government inspector. The rejected production model shall be corrected, where applicable, and shall become part of the first production lot.

4.3 Inspection provisions.

4.3.1 Inspection of components.- Inspection requirements for component parts and subassemblies are contained herein and on SQAPS listed on LSQAP-8244840.

4.3.1.1 The contents of the SQAP shall represent the minimum inspection requirements and shall be the basis for acceptance or rejection by lot of those components covered by the SQAP. The Government reserves the right, however, to inspect for the requirements of all drawings and specifications, and to reject individual non-conforming items.

4.3.1.2 Contractor's inspection for all components not covered by a SQAP shall be in accordance with the inspection procedure of the written inspection plan required by 4.1.1. The plan shall include procedures for checking all positional and functional characteristics shown on the drawings which indicate precision requirements that have a bearing on the functional accuracy and performance of the instrument.

4.3.2 Control of materials.- Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or subcontractor's facilities, or in other suitably equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.



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4.3.3 Lot formation.- Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.3.3.1 Acceptance tests.- The classification of defects in Table I shall constitute the minimum inspection requirements to be performed by the contractor prior to acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

4.3.3.2 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lot shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

TABLE I - CLASSIFICATION OF DEFECTS

For Normal Inspection use Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

For Reduced Inspection use Level I, and Sampling Plan Table IV-A of Standard MIL-STD-105

AQL value is expressed in percent defective.

| MAJOR: AQL 1.0% | Req't Par. | QA Par. | Test Method |
|---|---|---|---|
| 1. Orientation | 3.6.1 | 4.4.1.1 | Device, Special Testing (DST) |
| 2. Level vials | 3.6.2 | 4.4.1.2 | DST |
| 3. Horizontal alinement | 3.6.3 | 4.4.1.3 | DST |
| 4. Elevation jump | 3.6.4 | 4.4.1.4 | DST |
| 5. Telescope position | 3.6.5 | 4.4.1.5 | DST |
| 6. Identification marking | 3.6.6 | 4.4.2 | Visual |

4.3.4 Inspection equipment.

4.3.4.1 Contractor provisions.- Unless otherwise specified in the contract, the contractor shall provide and properly utilize all gaging, measuring and testing equipment of the required accuracy for the performance of all inspection examinations and tests required to be performed by the contractor under this contract. The contractor shall be responsible for the maintenance of the equipment including surveillance, calibration, and rehabilitation, on a regularly scheduled basis, and shall have available, all calibration equipment and other instrumentation essential for calibration and certification of the inspection equipment. The contractor also shall maintain records of inspection equipment surveillance and calibration, and shall make these records available for review by the Government. The Government reserves the right to certify inspection equipment and to use it for Government verification inspection or acceptance inspection.



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4.3.4.2 Government provisions.- Test equipment listed in IEL-8244840, when available, will be used for Government product inspection (reference 4.1.2(b)), and to settle disputes (reference 4.1.1). Drawing of this equipment will be furnished the contractor for informational purposes.

4.3.4.3 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes, established by product requirements, shall be capable of measuring these values within an accuracy of ten percent of the specified tolerance. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.3.4.3.1 Accuracy requirements for device, special testing.- The device for testing the performance requirements of the sightunit shall be equipped with suitable means for mounting the sightunit as specified. Provisions shall be made on the device to permit readings and settings of the sightunit throughout the full excursion of 6400 mils. All special set-up or calibration equipment necessary for certification of device accuracy shall be furnished by the contractor.

4.3.4.4 Performance tests are accomplished by means of an azimuth test fixture, perpendicularity gage, and an adapter. These fixtures and gages are depicted on drawings F7691596, C8227033, and C7651341, respectively and listed on IEL-8244840.

4.4 Test methods and procedures.

4.4.1 Performance.

4.4.1.1 Orientation.- The sightunit shall be mounted on a special testing device capable of positioning the dovetail axis and the dovetail flat in the vertical planes as specified in 3.6.1.

4.4.1.2 Level vials.- The sighunit shall be positioned with the elevation level vial bubble centered with respect to the vial graduations. The position of the cross level vial bubble shall be observed and shall be within the limits specified in 3.6.2.

4.4.1.3 Horizontal alinement.- The sightunit shall be positioned as specified in 4.4.1.1. The adapter indexes shall be set at zero. The line of sight through the telescope shall be directed at the collimator reticle. The telescope's horizontal reticle line shall be coincident with the collimator's horizontal reticle line. The adapter shall be rotated in a horizontal plane by means of the azimuth knob, a full excursion of 6400 mils. The telescope's horizontal reticle line shall not deviate more than plus or minus 1.7 mils as measured on the collimator reticle. This test shall be repeated at a minimum of four (4) azimuth positions.



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4.4.1.4 Elevation jump.- The azimuth rotation shall be reversed. The position of the telescope's horizontal line shall be observed and shall not exceed 0.5 mil over the values noted in 4.4.1.3. This test shall be repeated at a minimum of four (4) azimuth positions.

4.4.1.5 Telescope position.- The sightunit shall be positioned on a special testing device as specified in 3.6.5. The index of the telescope shall be adjusted to coincide with the index on the holder assembly. The horizontal reticle line of the telescope shall be observed with respect to the collimator's horizontal line and shall be within the one (1) degree limits as measured by the means provided on the collimator reticle.

4.4.2 Identification marking.- The identification marking shall be visually inspected for compliance with Standard MIL-STD-130.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Specification MIL-S-12214.

6. NOTES

6.1 Intended use.- Sightunits of the M34 Series are primarily intended for sighting of Recoiless rifles, rocket launchers, and mortars for direct or indirect fire in both elevation and azimuth.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of an application level of preservation, packaging and packing.
(c) Applicable stock number.
(d) Provisions for production model including date of delivery (see 3.1)

6.3 Design information.- Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon the specific request of the contractor.

6.4 Definitions.

6.4.1 Device, special testing (DST).- Special testing devices are defined as special inspection fixtures, instruments, and modified standard measuring equipment.



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6.4.2 SQAP (Supplementary Quality Assurance Provisions), LSQAP (List of SQAP's), and IEL (Inspection Equipment List).- Lists pertinent to each component and subassembly are identified by the Ordnance drawing number of the item prefixed by the above appropriate letter designation.

6.5 Packaging.- Specification MIL-S-12214, is also applicable to Sightunit, M34A2C since the only difference between Sightunit M34A2 and M34A2C is a change in the reticle pattern of the M62 Series Telescope. There are no external physical variations.

6.6 International standardization agreement.- Provisions of this specification are the subject of international standardization agreement. When amendment, revision, or cancellation of this specification is proposed the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:

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Preparing Activity:

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International interest (see 6.5)



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